

CUTTING TOOLS

DIAMOND DRILLS

Diamond is a solid form of carbon (an element).

An element is a substance made up of only one type of atom. Natural diamonds formed billions of years ago, under high pressures and temperatures deep within Earth, but nowadays they can also be made synthetically. Although it is best known as a shiny gemstone, diamond is the hardest naturally occurring material, a property which gives it a range of uses in industry and manufacturing.

CUT DIAMONDS

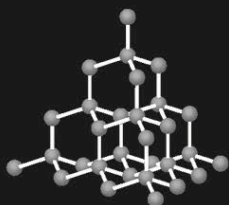
Diamonds have always been valued as gemstones. Rough stones are shaped and carefully cut to create many tiny surfaces called facets. When light enters the stone, these facets cause it to bounce around and split into different colors, creating a sparkly effect.



FORMS OF CARBON

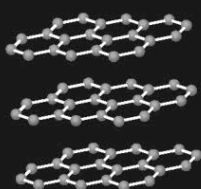
Pure carbon atoms can join together in different ways. The two best-known forms are diamond and graphite, but in 1985, a third type was discovered, called buckminsterfullerene. Their different structures give these substances unique properties. Graphite is soft and often used in pencils, while diamond is so hard it can cut through concrete.

DIAMOND



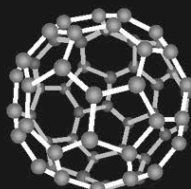
Atoms in diamond form a strong pyramid shape.

GRAPHITE



In graphite, atoms are arranged in layers that slide over each other.

BUCKMINSTERFULLERENE



Sixty carbon atoms are arranged in a ball in this form of carbon.

Diamond drill

Because of its hardness, diamond is used in many tools. In this image, a drill tipped with diamond polishes blocks of metal. Diamond is also added to the blades of saws in order to cut through tough materials such as steel.

